Bankrate

BI Manager Case Study

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Part 1:
BI and Data Infrastructure

Bringing the data together in Databricks and Looker

Source tables

users
sessions
partners
applications
mortgage_clickouts
clickouts
mortgage_revenue
revenue

Tables have foreign keys for clean joins to one another. Some tables, such as sessions and clickouts, may require **custom keys** to link.

There is opportunity to **centralize revenue and clickouts data** in Databricks for cleaner modeling, rather than relying on split tables.

Upstream transforms

Create **fct_revenue** table (**mortgage_revenue** and **revenue)** and **fct_clickouts** table (**mortgage_clickouts** and **clickouts)**

- product_type will differentiate the source

Create foreign key in **clickouts** table by deriving a *session_id* from *session_start_time* and *clickout_time*

Generally, complex transformations will take place upstream, while aggregations and custom measures will take place in Looker

Looker modeling

1. Import tables into Looker as fact and dimension views
   - dim_users.view.lkml
   - dim_partners.view.lkml
   - fct_sessions.view.lkml
   - fct_applications.view.lkml
   - fct_clickouts.view.lkml
   - fct_revenue.view.lkml

2. Create Explores in Model for specific context, limiting to 3-5 views in each
   - Revenue
   - Clickout Funnel
   - Applications
   - Sessions

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What users see and what they don’t

Visible to users

- Most dimensions and measures should be accessible, but this should be intentional to limit sprawl and maintain security
- IDs necessary for granular detail (e.g. application_id)
- FICO and income shown as grouped buckets
- Clear classifiers (e.g. "Approved / Declined" instead of 1/0)

Not visible to users

- IDs and keys that are only necessary for linking data
- IDs that are superseded by a name (e.g. partner_name)
- PII and sensitive data, such as income and FICO (permissioned access only)
- Unclear boolean or 1/0 measures

Leadership KPI Dashboard

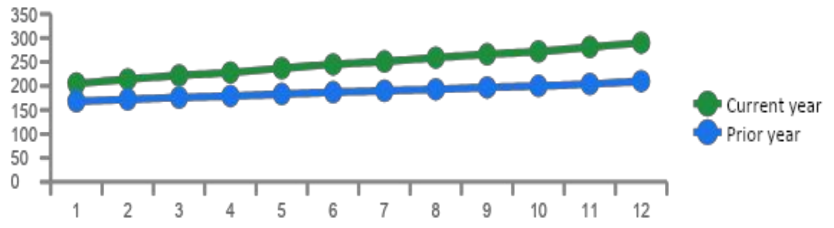
Partner ▾ | Channel ▾ | FICO Score Group ▾

Total Revenue (TTM)		Approvals (TTM)		Approval Rate (TTM)		Revenue / Application (TTM)	
$2.64M	YoY +8.2%	10,976	YoY +6.5%	11.0%	YoY +1.1%	$26.42	YoY +1.6%
	QoQ +3.1%		QoQ +2.4%		QoQ +0.5%		QoQ +0.7%
	MoM +1.4%		MoM +0.9%		MoM +0.2%		MoM +0.4%

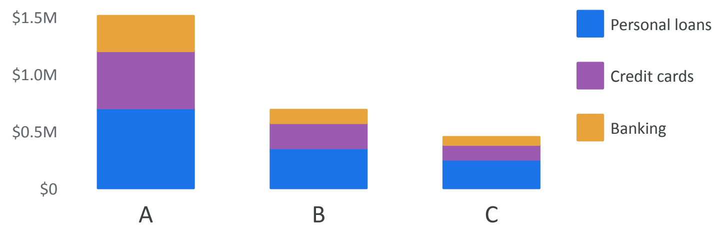
Acquisition Funnel

		Total
Sessions		1,200,000
Clickouts	35%	420,000
Applications	24%	100,000
Approvals	11%	10,976
Revenue	$241 per Application	$2.64M

Revenue Trend (YoY)



Revenue by Partner & Product



Core Aggregate Views

Executive Leadership

Focuses on overall performance and business health to determine go-forward strategy

- Revenue Fact Explore
- Acquisition Funnel Explore

Growth & Marketing

Focuses on acquiring customers, increasing applications and revenue, and marketing campaigns

- Acquisition Funnel Explore
- Application Explore
- Session Explore
- Marketing Campaign Explore

Partner/Revenue Management

Focuses on partner performance and revenue streams to inform relationships and deals and prioritize customer flow

- Revenue Fact Explore
- Partner Explore

Part 2:
Partner Matching Analysis

Partner funnel overview

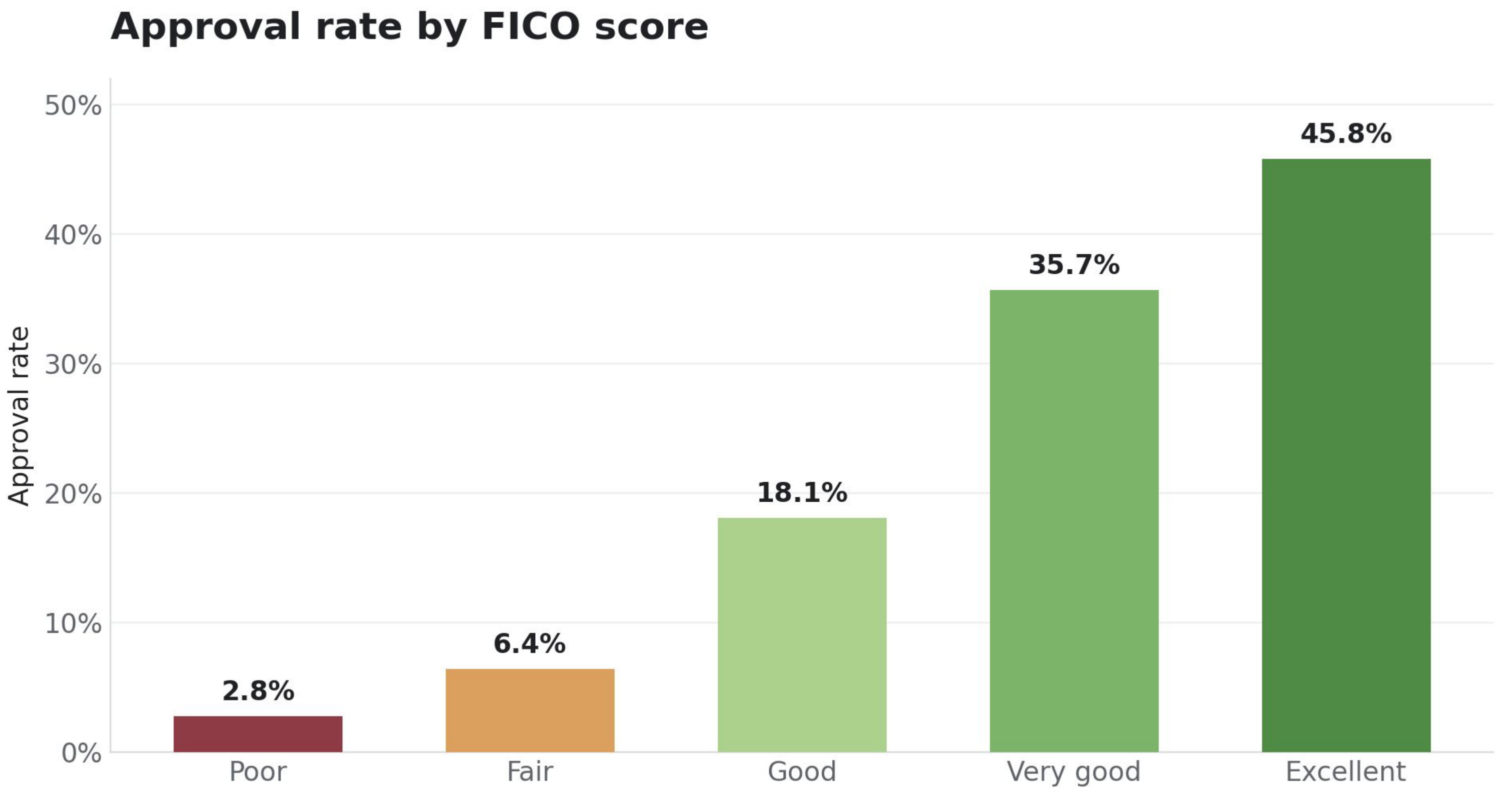
Hypothesis: The current customer-to-partner matching process is suboptimal. There is an opportunity to maximize revenue by better pairing customers with partners and increasing approvals, especially given customer data, approval rates, and payouts per loan.

Partner	Applications	Payout per Approval	Approval Rate	Revenue	Revenue per Application
A	55,000 (55%)	$250	11.0%	$1,507,750	**$27.41**
B	27,500 (27%)	$350	7.1%	$686,000	**$24.95**
C	17,500 (18%)	$150	17.1%	$447,750	**$25.59**
Total	100,000 (100%)	—	11.0%	$2,641,500	$26.42

Partner A has the **most applications** and **highest RPA** compared to other partners, but sits in the middle in terms of payout and approval rate.
Partner B has roughly a quarter of all applications and the **highest payout**, but the **lowest approval rate and lowest RPA**.
Partner C has the fewest applications and **lowest payout**, but the **highest approval rate**.

In order to determine how to optimally match customers to partners, we need to better understand the customer data.

What determines approvability?



Housing Ratio

Ideal (<= 28%) represented in 48% of applications with a 15% approval rate

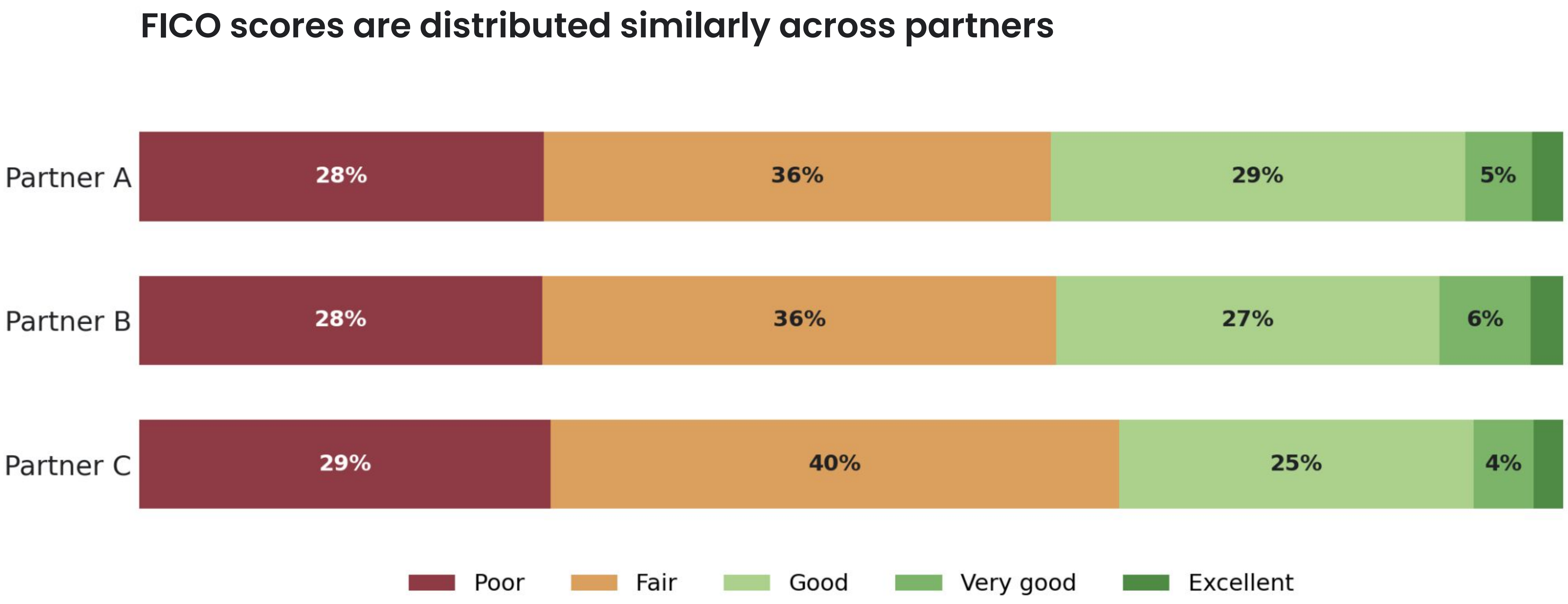
Prior Bankruptcy or Foreclosure

97% of applications do not have prior bankruptcy or foreclosure

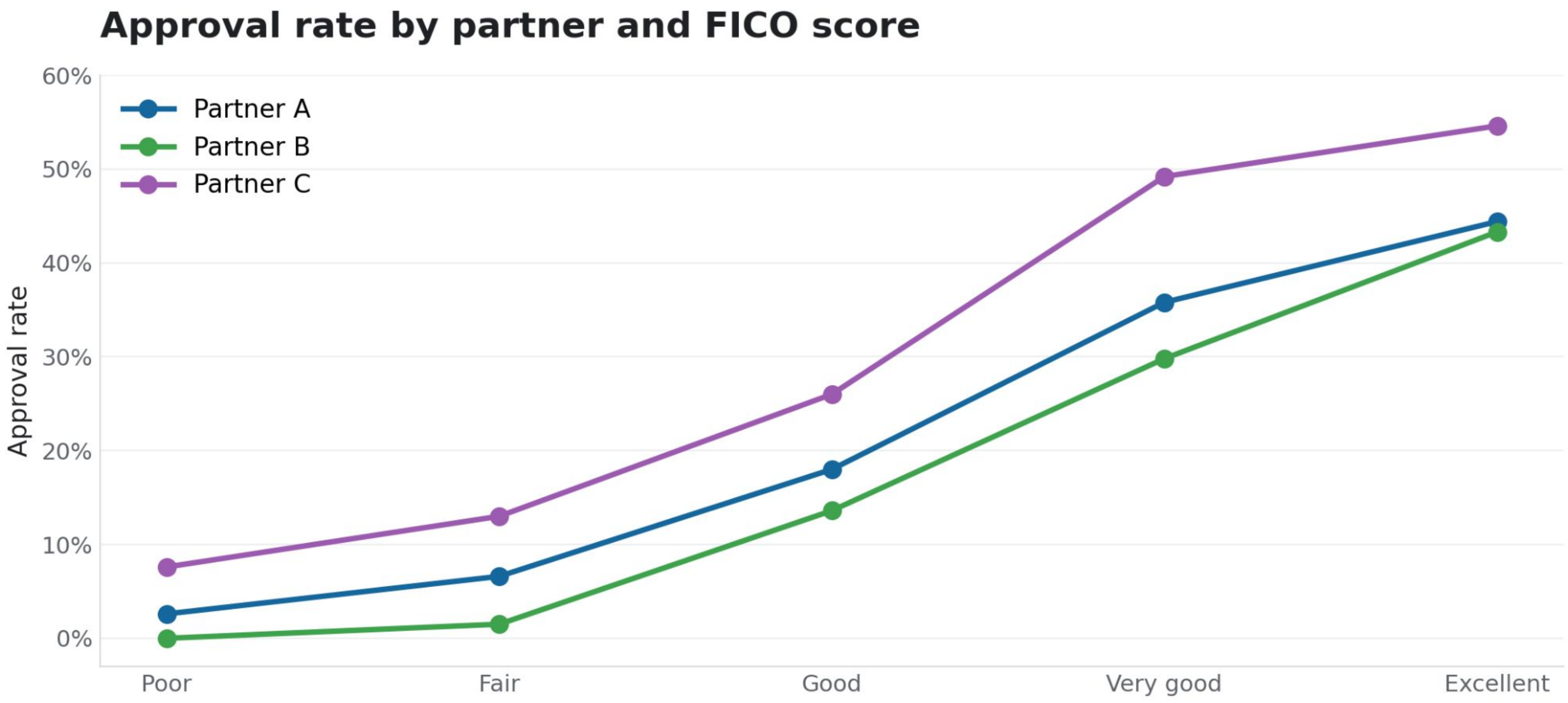
Employment

77% of customers are FTE with 12% approval rate

Partners and FICO score distribution



Approval rates differ across partners and FICO



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Expected values for FICO scores and partners

The payouts for each partner (A = $250, B = $350, C = $150) and the approval rates give us the expected value for each FICO score group. Partner B has the most expected value for Good, Very Good, and Excellent scores, while Partner C has the most expected value for Poor and Fair.

Expected Value = (Approval Rate x Payout)

Partner A *never* has the best expected value, but it receives the majority of applications. This is a reliable partner and solid fallback, and the high volume allows for routing customers at the margins to Partner B and C to maximize revenue opportunities.

FICO Score	Applications	Partner A Expected Value	Partner B Expected Value	Partner C Expected Value
Excellent	2,188	$111	**$151.55**	$81.90
Very Good	5,102	$89.50	**$104.30**	$73.80
Good	27,760	$45	**$47.60**	$39
Fair	36,475	$16.5	$5.25	**$19.50**
Poor	28,475	$6.5	$0	**$11.40**

Recommendations

- Match **lower FICO scores** with **Partner C,** which approves **2x** as often as Partner A, and avoid matching with Partner B.
- Match **higher FICO scores** with **Partner B** to maximize expected revenue from more valuable applications.
- **Partner A**, which receives 55% of applications, provides meaningful **volume to redistribute** based on **expected value**
- Test and **validate updated routing** against a subset of customers and refine model before full deployment.